

LAPORAN WORKSHOP KECERDASAN BUATAN

PERTEMUAN 2



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I. Materi

Lingkungan virtual (*virtual environment*) Python sangat dibutuhkan saat Anda mulai mengembangkan program yang kompleks menggunakan pustaka pihak ketiga. Proyek yang berbeda sering kali bergantung pada versi perangkat lunak yang berbeda pula (misalnya TensorFlow versi 1 dan 2), sehingga sangat rentan memicu konflik versi pada sistem Anda jika disatukan.

Untuk mengatasi masalah ini, Anda dapat membuat lingkungan Python yang terisolasi untuk masing-masing proyek, sehingga berbagai versi pustaka dapat berjalan berdampingan di satu komputer tanpa bentrok. Lingkungan ini bisa dibuat menggunakan alat seperti Anaconda atau Virtualenv—yang ukurannya lebih kecil dan mudah digunakan—dengan panduan instalasi lengkap yang bisa Anda lihat pada Lampiran D atau situs resmi Virtualenv.

Buku ini menggunakan Python IDLE, serta kombinasi platform lain seperti Jupyter, Kaggle, dan Google Colab. Meskipun bukan buku panduan khusus tentang pemrograman Python secara mendalam, buku ini tetap memberikan pengantar dasar yang singkat dan sederhana sebagai pelengkap materi.

Seperti mempelajari bahasa pemrograman pada umumnya, langkah pertama dimulai dengan program "Hello World" untuk memahami konsep *standard output* (tampilan layar) dan *standard input* (masukan keyboard). Setelah menguasai dasar tersebut, proses belajar akan dilanjutkan ke konsep yang lebih terstruktur seperti variabel, perulangan (looping), percabangan (if-else), hingga fungsi.

II. Alat & Bahan

- A. Laptop
- B. Microsoft Word
- C. Visual Studio Code

III. Tugas

A. TUGAS PENDAHULUAN

1. EXAMPLE 2.1: THE HELLO.PY PROGRAM

```
#Example 2.1
print("Hello World!")
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> &
Hello World!
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

2. EXAMPLE 2.2: THE HELLO2.PY PROGRAM

```
#Example 2.2
'''
This is a multiple line
comments.
```

```
'''  
print("Hello World")
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> &  
Hello World  
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

3. EXAMPLE 2.3: THE HELLO3.PY PROGRAM

```
#Example 2.3  
a = 5  
b = 7  
c = a + b  
print("c = " , c)  
d = 'x'  
e = 'y'  
print(d + e)  
f = "Hello"  
g = "World"  
l = f + " " + g  
print(l)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> &  
c = 12  
xy  
Hello World  
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

4. EXAMPLE 2.4: THE LOOP.PY PROGRAM

```
#Example 2.4  
for i in range(5):  
    print(i)  
print("Finished")
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> &  
0  
1  
2  
3  
4  
Finished  
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

5. EXAMPLE 2.5: THE LOOP2.PY PROGRAM

```
#Example 2.5  
for i in range(5):  
    for j in range(5):  
        x = i * j
```

```

        print(x)
        print("Inner Loop Finished")
    print("Outer Loop Finished")

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_5.py"
0
Inner Loop Finished
4
Inner Loop Finished
8
Inner Loop Finished
12
Inner Loop Finished
16
Inner Loop Finished
Outer Loop Finished
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

6. EXAMPLE 2.6: THE LOOP3.PY PROGRAM

```

#Example 2.6
x = 0
while x < 5:
    x = x + 1
    print(x)
print("Finished")

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_6.py"
1
2
3
4
5
Finished
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

7. EXAMPLE 2.7: THE IFELSE.PY PROGRAM

```

#Example 2.7
x = 60
if x >= 70:
    print("Excellent")
elif x >= 60:
    print("Good")
else:
    print("OK")

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_7.py"
Good
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

8. EXAMPLE 2.8: THE FUNCS.PY PROGRAM

```

#Example 2.8
def add (x,y):
    z = x + y
    return z
t = add(30, 20)
print(t)

```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_8.py"
50
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

9. EXAMPLE 2.9: THE ARRAY.PY PROGRAM

```
#Example 2.9
def maxarray (xs):
    m = xs[0]
    for x in xs:
        if m < x:
            m = x
    return m
data = [0,1,2,3,4,5]
t = maxarray(data)
print(t)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_9.py"
5
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

a. EXERCISE 2.1

Modify the Python program from Example 2.9 to create a function called `minarray(xs)` to find out the minimum value of an array

```
def minarray(xs):
    m = xs[0]
    for x in xs:
        if m > x:
            m = x
    return m

data = [0, 1, 2, 3, 4, 5]
t = minarray(data)
print(t)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_9e1.py"
0
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

b. EXERCISE 2.2

Modify the Python program from Example 2.9 to create a function called `sortarray(xs)` to sort the array in ascending order.

```
def sortarray(xs):
    n = len(xs)
    # Melakukan perulangan untuk
    membandingkan setiap elemen
    for i in range(n):
        for j in range(0, n - i - 1):
```

```

        # Jika elemen saat ini lebih
        besar dari elemen berikutnya, tukar
        posisinya
        if xs[j] > xs[j+1]:
            xs[j], xs[j+1] = xs[j+1],
            xs[j]
        return xs

# Mengacak urutan data untuk membuktikan
fungsi berjalan dengan benar
data = [5, 2, 0, 4, 1, 3]
t = sortarray(data)
print(t)

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_9e2.py"
[0, 1, 2, 3, 4, 5]
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

10. EXAMPLE 2.10: THE INPUT.PY PROGRAM

```

#Example 2.10
print('What is your name? ')
x = input()
print('Hello ' + x + "!")

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_10.py"
What is your name?
Tony
Hello Tony!
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

11. EXAMPLE 2.11: THE INPUT2.PY PROGRAM

```

#Example 2.11
print('Input a number: ')
x = input()
y = int(x)
y = y ** 2
print('The square of ' + str(x) + " is " + str(y))

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_11.py"
Input a number:
110
The square of 110 is 12100
PS D:\Tugas\Semester 5\AI\AI_P1C2>

```

a. EXERCISE 2.3

Modify the Python programs from Example 2.11 so that it reads a float number from the keyboard. You can also use `float()` to convert text to float numbers, which are the numbers with decimal points. Then calculate the square and display it on the screen.

```

print('Input a number: ')

```

```
x = input()
y = float(x)
y = y ** 2
print('The square of ' + str(x) + " is " +
str(y))
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_11e1.py"
Input a number:
20
The square of 20 is 400.0
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

b. EXERCISE 2.4

Based on the Python programs in Examples 2.10 and 2.11, write a Python program that reads the name, the age, and the marks from the keyboard. The age should be read as an integer, and marks should be read as a float number. Then display them on the screen.

```
# Membaca nama (sebagai teks/string)
print('What is your name? ')
name = input()

# Membaca umur (dan mengubahnya menjadi
integer)
print('What is your age? ')
age_str = input()
age = int(age_str)

# Membaca nilai/marks (dan mengubahnya
menjadi float)
print('What are your marks? ')
marks_str = input()
marks = float(marks_str)

# Menampilkan hasilnya ke layar
print('Hello ' + name + "!")
print('Age: ' + str(age))
print('Marks: ' + str(marks))
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_11e2.py"
What is your name?
Dandi
What is your age?
20
What are your marks?
1
Hello Dandi!
Age: 20
Marks: 1.0
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

12. EXAMPLE 2.12: THE INPUTLIST.PY PROGRAM

```
#Example 2.12
```

```
s = input("Input a list of numbers: ")
numbers = list(map(int, s.split()))
print(numbers)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_12.py"
Input a list of numbers: 1 2 3 4 5
[1, 2, 3, 4, 5]
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

13. EXAMPLE 2.13: THE MATH.PY PROGRAM

```
#Example 2.13
import math
r = 5
cir = 2 * math.pi * r
print('The circumference is ' + str(cir))
x = 5
y = math.sqrt(x)
print("The square root of " + str(x) + " is " + str(y))
x = 7
y = math.factorial(x)
print("The factorial of " + str(x) + " is " + str(y))
x = 16.4
y = math.floor(x)
print("The floor of " + str(x) + " is " + str(y))
x = 16.4
y = math.ceil(x)
print("The ceiling of " + str(x) + " is " + str(y))
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_13.py"
The circumference is 31.41592653589793
The square root of 5 is 2.23606797749979
The factorial of 7 is 5040
The floor of 16.4 is 16
The ceiling of 16.4 is 17
PS D:\Tugas\Semester 5\AI\AI_P1C2> █
```

a. EXERCISE 2.5

Based on the Python programs from Examples 2.10 and 2.13, write a Python program that reads a list of float numbers from the keyboard, then uses the `math.sin()` function to calculate the sine value of the numbers, and finally displays the results on the screen.

```
import math

# Meminta pengguna memasukkan daftar angka
(dipisahkan oleh spasi)
```

```

print('Input a list of float numbers
(separated by space): ')
user_input = input()

# Memecah teks input menjadi list string
berdasarkan spasi
number_strings = user_input.split()

# Melakukan perulangan untuk setiap angka
yang dimasukkan
for num_str in number_strings:
    # Mengubah teks menjadi angka desimal
    (float)
    x = float(num_str)

    # Menghitung nilai sinus menggunakan
    math.sin()
    y = math.sin(x)

    # Menampilkan hasil ke layar
    print("The sine of " + str(x) + " is "
+ str(y))

```

```

PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_13e1.py"
Input a list of float numbers (separated by space):
1.2 1.3 2.5 5.6
The sine of 1.2 is 0.9320390859672263
The sine of 1.3 is 0.963558185417193
The sine of 2.5 is 0.5984721441039564
The sine of 5.6 is -0.6312666378723216
PS D:\Tugas\Semester 5\AI\AI_P1C2>

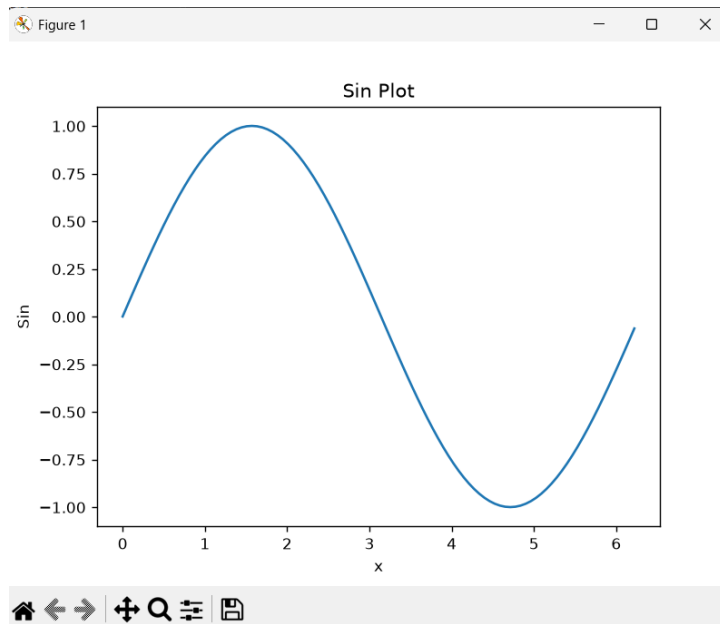
```

14. EXAMPLE 2.14: THE PLOT.PY PROGRAM

```

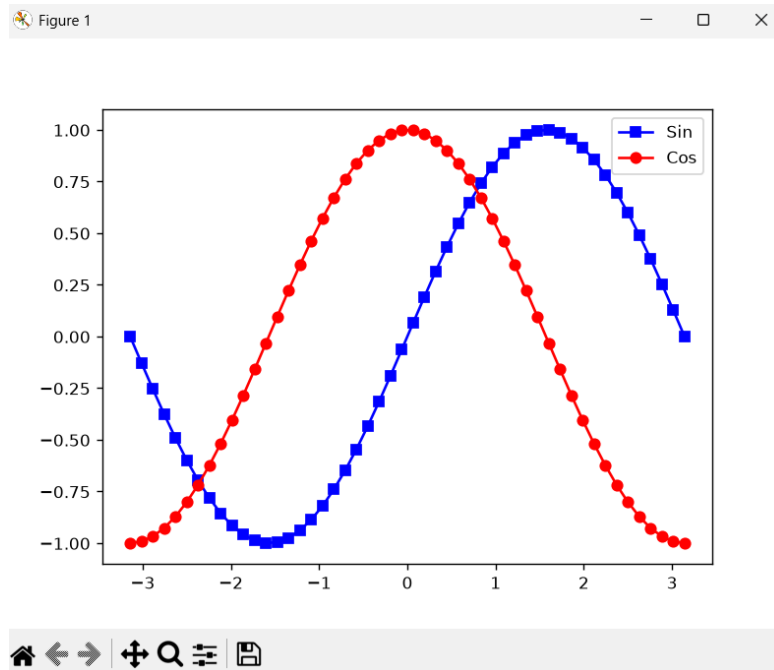
#Example 2.14
import matplotlib.pyplot as plt
import numpy as np
x = [i for i in range(100)]
x = np.array(x)
x = 2 * np.pi * x * 0.01
y = np.sin(x)
plt.plot(x,y)
plt.title('Sin Plot')
plt.xlabel('x')
plt.ylabel('Sin')
plt.show()

```



15. EXAMPLE 2.15: THE PLOT2.PY PROGRAM

```
#Example 2.15
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-np.pi, np.pi, 50)
y1 = np.sin(x)
y2 = np.cos(x)
plt.plot(x, y1, color = 'blue', marker = "s",
label='Sin')
plt.plot(x, y2, color = 'red', marker = "o",
label='Cos')
plt.legend()
plt.show()
```



a. EXERCISE 2.6

Based on the Python programs from Examples 2.14 and 2.15, write a Python program that plots different math functions, such as $y = 3x + 4$ and $y = 2x^2 + 1$ and $y = x^3 + 9$ on the same graph using different colors and also displays the legend.

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Membuat rentang nilai x, misalnya dari
-10 hingga 10 sebanyak 100 titik
x = np.linspace(-10, 10, 100)

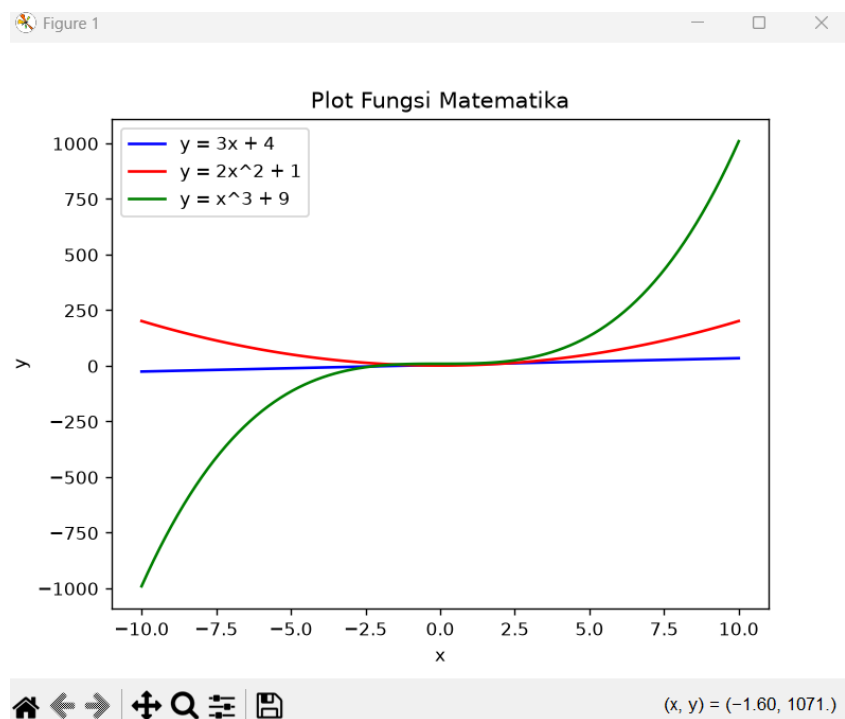
# 2. Mendefinisikan persamaan fungsi
matematika
y1 = 3 * x + 4
y2 = 2 * x**2 + 1
y3 = x**3 + 9

# 3. Menggambar plot untuk setiap fungsi
dengan warna dan label yang berbeda
plt.plot(x, y1, color='blue', label='y = 3x
+ 4')
plt.plot(x, y2, color='red', label='y =
2x^2 + 1')
plt.plot(x, y3, color='green', label='y =
x^3 + 9')
```

```
# 4. Menambahkan judul dan label sumbu
plt.title('Plot Fungsi Matematika')
plt.xlabel('x')
plt.ylabel('y')

# 5. Menampilkan legenda grafik
plt.legend()

# 6. Menampilkan grafik ke layar
plt.show()
```



16. EXAMPLE 2.16: THE DATA.PY PROGRAM

```
#Example 2.16
import pandas as pd
data = {'Name':
['Tony', 'Robert', 'John', 'Alice'],
'Age': [18, 24, 19, 21],
}
df = pd.DataFrame (data, columns =
['Name', 'Age'])
print (df)
df.to_csv('test.csv', index=False)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_16.py"
   Name Age
0    Tony  18
1  Robert  24
2    John  19
3    Alice 21
PS D:\Tugas\Semester 5\AI\AI_P1C2>
```

17. EXAMPLE 2.17: THE DATA2.PY PROGRAM

```
#Example 2.17
import pandas
df = pandas.read_csv('test.csv')
print (df)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_17.py"
   Name Age
0    Tony  18
1  Robert  24
2    John  19
3    Alice 21
PS D:\Tugas\Semester 5\AI\AI_P1C2>
```

18. EXAMPLE 2.18: THE EXCEL.PY PROGRAM

```
#Example 2.18
#pip install xlswriter
import pandas as pd
data = {'Name':
['Tony', 'Robert', 'John', 'Alice'],
'Age': [18, 24, 19, 21],
}
df = pd.DataFrame (data, columns =
['Name', 'Age'])
print (df)
writer = pd.ExcelWriter("test.xlsx",
engine='xlswriter')
#df.to_excel(writer, sheet_name='Sheet1',
# startrow=7, startcol=4, header=False,
index=False)
df.to_excel(writer, sheet_name = 'Sheet1',
index=False)
writer.close()
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_18.py"
   Name Age
0    Tony  18
1  Robert  24
2    John  19
3    Alice 21
PS D:\Tugas\Semester 5\AI\AI_P1C2>
```

The screenshot shows the Microsoft Excel interface. The 'Home' tab is active. The ribbon includes 'File', 'Home', 'Insert', 'Draw', 'Page Layout', 'Formulas', and 'Data'. The 'Clipboard' group shows 'Cut', 'Copy', 'Paste', and 'Format Painter'. The 'Font' group shows 'Calibri' font, size '11', and options for bold, italic, underline, and text color. The 'Name' box shows 'A1'. The spreadsheet has two columns: 'Name' and 'Age'. The data is as follows:

	Name	Age					
1	Tony	18					
2	Robert	24					
3	John	19					
4	Alice	21					

19. EXAMPLE 2.19: THE EXCEL2.PY PROGRAM

```
#Example 2.19
#pip install xlswriter
import pandas
df = pandas.read_excel('test.xlsx',
sheet_name='Sheet1')
print(df)
```

```
PS D:\Tugas\Semester 5\AI\AI_P1C2> & C:\Python313\python.exe "d:/Tugas/Semester 5/AI/AI_P1C2/Source Code/P1C2_19.py"
   Name  Age
0  Tony   18
1 Robert   24
2  John   19
3  Alice   21
PS D:\Tugas\Semester 5\AI\AI_P1C2> 
```